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CASE REPORT : WHEN INVASIVE LOBULAR CARCINOMA MEETS
HIGH-GRADE DUCTAL CARCINOMA IN SITU,
WHAT DOES IT LOOK LIKE?



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INTRODUCTION

Both Invasive Lobular Carcinoma (ILC) and Ductal Carcinoma in Situ (DCIS) have a predilection for cellular growth along the ducts that likely explains imaging findings without discrete mass formation. Despite the similarity of growth patterns, the clinical courses of ILC and DCIS are different

CASE PRESENTATION

A 41-year-old female with a chief complaint of a mass in her left breast, approximately two years, progressively enlarged. A hard mass was palpable in the left breast region, with associated skin erythema

Breast ultrasound (US) revealed a solid, hypoechoic, amorphous mass in the left retro-papillary and retro-areolar regions extending to the upper quadrant. The mass measured approximately 6.28 × 3.05 cm, indistinct, irregular, and spiculated margins, accompanied by microcalcifications within the lesion. Elastography demonstrated a hard mass (Fig. 1 and 2). Mammography revealed heterogeneously dense breast tissue (BI-RADS composition C) with a hyperdense mass indistinct and mildly spiculated margins with pleomorphic heterogeneous calcifications in a segmental distribution, with minimal cutaneous thickening, alongside nipple retraction. No axillary lymphadenopathy was identified (Fig. 3 and 4). BI-RADS 5.

The breast biopsy specimen demonstrated invasive and carcinoma in situ components. Pathological Impression: Invasive lobular carcinoma, grade 2, accompanied by high-grade ductal carcinoma in situ. Immunohistochemistry (IHC): ER: Positive (moderate to strong intensity in 40% of tumor cells). PR: Negative (no staining observed). HER 2 : positive. Ki-67: Expressed (strong intensity in 60% of tumor cells; high proliferative index). Molecular subtypes of breast cancer is Luminal B

The patient is currently undergoing 8 cycles of AC-T neoadjuvant chemotherapy (doxorubicin (Adriamycin) and cyclophosphamide (AC) followed by a taxane (T)) and scheduled for modified radical mastectomy

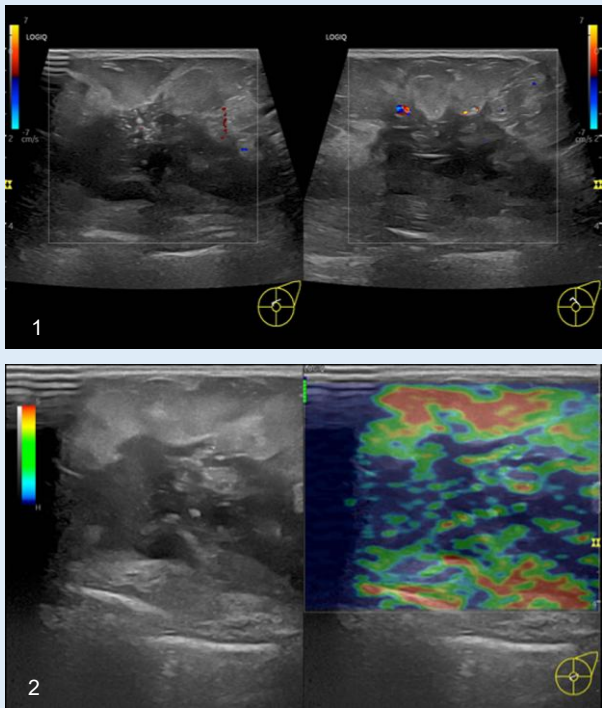


Figure 1 & 2 : US revealed a solid, hypoechoic, amorphous mass, with indistinct, irregular, and spiculated margins, accompanied by microcalcifications within the lesion. Elastography demonstrated a hard mass

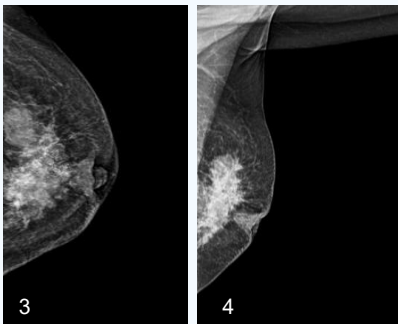


Figure 3 & 4 :Mammography revealed heterogeneously dense breast tissue (BI-RADS composition C) with a hyperdense mass indistinct and mildly spiculated margins with pleomorphic heterogeneous calcifications in a segmental distribution,with minimal cutaneous thickening, alongside nipple retraction. No axillary lymphadenopathy was identified

DISCUSSION

ILC frequently presents on mammography as a mass with spiculated or indistinct margins, typically without associated calcifications. Occasionally, imaging features can be subtle, manifesting only as architectural distortion, asymmetry, or focal shrinkage. Conversely, DCIS predominantly manifests as suspicious mammographic microcalcifications. The most common US patterns of ILC are a hypoechoic mass with angular or indistinct margins with or without posterior acoustic shadowing; an ill-defined area of altered inhomogeneous echotexture without identifiable margins; shadowing with no associated mass; or a circumscribed mass. Ultrasonography is not routinely used for DCIS evaluation.

In this patient, the morphology strongly suggests ILC while the prominent microcalcification increase the likelihood of an accompanying DCIS. Breast MRI was nor performed in this patient because the lesions’s appearance an boundaries were sufficiently clear on ultrasound and mammography.

E-cadherin staining was nor performed in this case; however majority of ductal carcinoma express cytoplasmic E-Cadherine, whereas most lobular carcinomas lack E-Cadherine expression.

CONCLUSION

ILC is more likely than DCIS to form a mass, meanwhile calcifications are very commonly found in DCIS. An MRI and examination of E-Cadherin may be considered to monitor the response following adjuvant chemotherapy and to confirm a more accurate diagnosis. The combination of imaging features with patient history and blood or tissue biomarkers would enhance the value and the diagnostic accuracy of imaging for breast cancer screening, diagnosis, prognosis, and treatment.

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